

Security Operations Architecture Topology

Splunk manager and monitored endpoints in the isolated UTM lab network.

Node Inventory

Node	IP address	System	Role	Telemetry / access
Splunk Manager	192.168.64.1	macOS host	Indexer, search head, receiver	Splunk Web :8000, UF input :9997, SignalScope :8765
Windows Endpoint	192.168.64.2	Windows 11 VM	Monitored endpoint	Security/System/Application/Sysmon via Splunk UF
Linux Endpoint	192.168.64.4	Kali Linux VM	Monitored endpoint + SSH test target	/var/log/auth.log and /var/log/syslog via Splunk UF
SOC Analyst	127.0.0.1:8765	Local browser	Live investigation console	SignalScope queries Splunk management API locally

Security Boundary

- The lab lives inside the UTM virtual network 192.168.64.0/24. Endpoint telemetry stays inside that network.
- Windows and Kali forward logs to the Splunk Manager on TCP 9997. The analyst views Splunk Web on TCP 8000.
- SignalScope is intentionally localhost-bound at 127.0.0.1:8765; another PC that clones the repo hosts its own local console.
- Attack simulations are controlled lab events used to produce realistic detection evidence, not internet-facing activity.

Figure 1 - Computer Node Topology

Literal connected devices and analyst/evidence nodes.

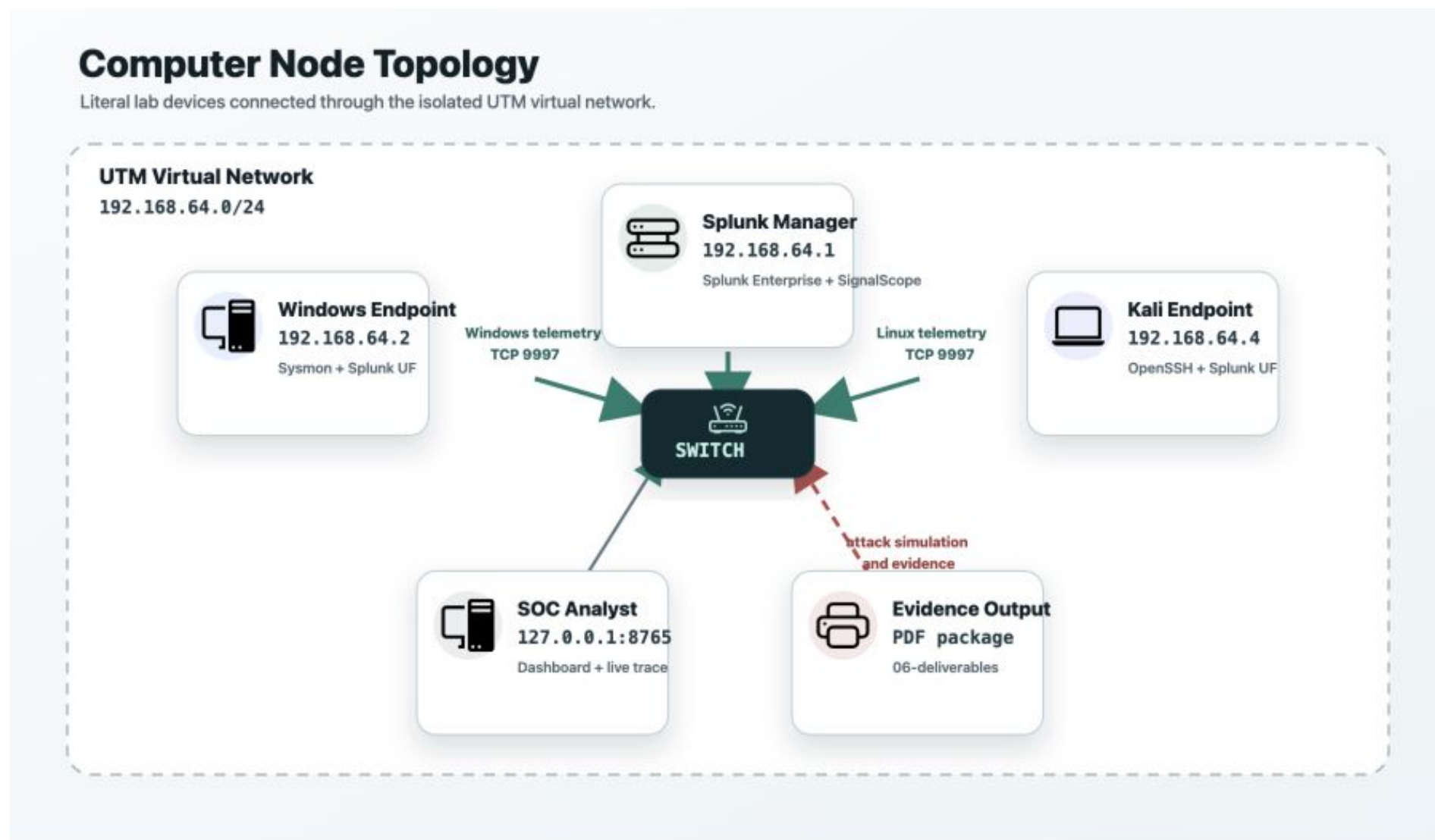


Figure 2 - Detailed Telemetry and Attack Flow

Splunk forwarding paths, local analyst access, and controlled attack simulation paths.

